IV B.Tech I Semester

23A31703c	**HEALTHCARE AI (Professional Elective-V)**	**L**	**T**	**P**	**C**
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Course Objectives:

- To provide a foundational understanding of AI applications in healthcare.
- To familiarize students with medical data types, preprocessing, and ethical considerations.
- To explore ML and DL algorithms tailored for diagnosis, prognosis, and treatment recommendations.
- To expose students to real-world healthcare systems and AI solutions like predictive modeling, EHRs, and medical imaging.
- To enable students to design, evaluate, and deploy AI-driven healthcare applications.

Course Outcomes:

After completing this course, students will be able to:

- Understand the scope, challenges, and benefits of AI in healthcare.
- Apply data preprocessing and modeling techniques specific to biomedical data.
- Analyze the performance of ML/DL models in clinical contexts.
- Develop AI-driven applications for tasks like disease diagnosis, drug discovery, and patient monitoring.
- Evaluate ethical, legal, and societal implications of AI in healthcare.

UNIT I: Introduction to AI in Healthcare

Overview of Healthcare Systems and Data Ecosystem, AI in Clinical Decision Support Systems (CDSS), Types of Medical Data: EHRs, Imaging, Genomic, Sensor Data, Applications of AI in Diagnosis, Prognosis, and Monitoring, Use Cases: Radiology, Pathology, Oncology, Cardiology, Limitations and Challenges of AI in Healthcare, AI for Telemedicine and Remote Patient Monitoring –

UNIT II: Medical Data Preprocessing and Feature Engineering

Data Cleaning, Imputation, and Normalization for Clinical Data ,Handling Missing Values, Outliers, and Bias ,Feature Engineering from EHRs and Time-Series Data ,Text Mining for Medical Notes using NLP ,Encoding Diagnosis and Procedure Codes (ICD, CPT) ,Temporal Pattern Extraction from Clinical Sequences ,Data Privacy, Anonymization, and HIPAA Compliance –

UNIT III: Machine Learning & Deep Learning in Healthcare

Supervised Learning for Risk Prediction and Classification ,Unsupervised Learning for Patient Segmentation ,Deep Learning for Medical Imaging: CNNs, Transfer Learning ,Recurrent Neural Networks for Time-series Clinical Data ,Survival Analysis and Time-to-Event Prediction ,Model Evaluation Metrics: Sensitivity, Specificity, AUC**,** Handling Imbalanced Datasets in Healthcare – **,** Interpretability in Medical ML Models (LIME, SHAP) –

UNIT IV: Specialized Healthcare AI Applications

AI for Disease Diagnosis: Diabetes, Cancer, Heart Disease ,AI in Medical Imaging: X-ray, MRI, CT Scan Analysis ,Predictive Modeling for ICU Admission & Mortality Risk ,AI in Genomics and Personalized Medicine ,Drug Discovery and Repurposing with AI ,Chatbots and Virtual Health

Assistants ,Remote Monitoring using IoT & Wearables + AI ,Designing an End-to-End AI Solution for a Medical Use Case –

UNIT V: Ethics, Regulation, and Future Directions in Healthcare AI

Ethical AI in Healthcare: Bias, Fairness, and Accountability ,Regulatory Landscape: FDA Approval, CE Marking – **,** Explainable AI and Clinical Trust – **,** Federated Learning for Privacy-Preserving AI ,Clinical Trials and AI Decision-Support Tools – **,** Case Studies: Google DeepMind, IBM Watson Health, PathAI ,Responsible Deployment of AI in Healthcare Settings – **,** Capstone: Building a Responsible Healthcare AI Proposal –

Textbooks:

1. Jiang, Fei et al. – Artificial Intelligence in Healthcare: Past, Present and Future.
2. Kevin Frick – Introduction to Healthcare AI.
3. Eric Topol – Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again.

Reference Books:

1. Mathias Goyen – AI in Medical Imaging.
2. Bertalan Meskó – The Guide to the Future of Medicine: Technology and The Human Touch.
3. Peter Szolovits – Artificial Intelligence in Medicine (Morgan Kaufmann).

Online Learning Resources:

- Coursera: AI for Medicine Specialization (offered by DeepLearning.AI) https://www.coursera.org/specializations/ai-for-medicine
- HarvardX: Data Science in Healthcare (edX) https://online-learning.harvard.edu/course/data-science-healthcare